

雅可比矩阵构建 (微分变换法)



雅可比矩阵构建（微分变换法）

$$\begin{bmatrix} v_x \\ v_y \\ v_z \\ \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} J_1 & J_2 & \dots & J_i & \dots & J_n \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \dot{q}_n \end{bmatrix}$$

雅克比矩阵

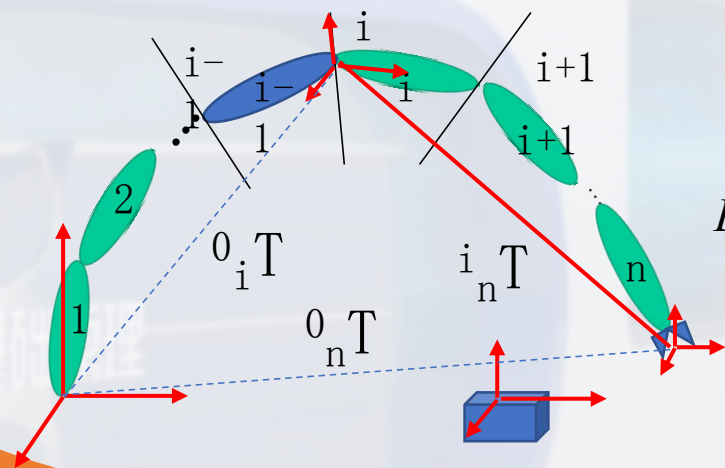


雅可比矩阵构建（微分变换法）

$$\begin{bmatrix} {}^T d_x \\ {}^T d_y \\ {}^T d_z \\ {}^T \delta_x \\ {}^T \delta_y \\ {}^T \delta_z \end{bmatrix} = \begin{bmatrix} n_x & n_y & n_z & (p \times n)_x & (p \times n)_y & (p \times n)_z \\ o_x & o_y & o_z & (p \times o)_x & (p \times o)_y & (p \times o)_z \\ a_x & a_y & a_z & (p \times a)_x & (p \times a)_y & (p \times a)_z \\ 0 & 0 & 0 & n_x & n_y & n_z \\ 0 & 0 & 0 & o_x & o_y & o_z \\ 0 & 0 & 0 & a_x & a_y & a_z \end{bmatrix} \begin{bmatrix} d_x \\ d_y \\ d_z \\ \delta_x \\ \delta_y \\ \delta_z \end{bmatrix}$$



微分变换关系式



$$D = \begin{bmatrix} d_x \\ d_y \\ d_z \\ \delta_x \\ \delta_y \\ \delta_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ d_z \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad D = \begin{bmatrix} d_x \\ d_y \\ d_z \\ \delta_x \\ \delta_y \\ \delta_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \delta_z \end{bmatrix}$$



雅克比矩阵构建(第i列)

✓ 对于转动关节，雅克比矩阵的第 i 列计算式为：

$${}^T J_i = \begin{bmatrix} (P \times n)_z \\ (P \times o)_z \\ (P \times a)_z \\ n_z \\ o_z \\ a_z \end{bmatrix} \begin{bmatrix} {}^T d_x \\ {}^T d_y \\ {}^T d_z \\ {}^T \delta_x \\ {}^T \delta_y \\ {}^T \delta_z \end{bmatrix} = \begin{bmatrix} n_x & n_y & n_z & (p \times n)_x & (p \times n)_y & (p \times n)_z \\ o_x & o_y & o_z & (p \times o)_x & (p \times o)_y & (p \times o)_z \\ a_x & a_y & a_z & (p \times a)_x & (p \times a)_y & (p \times a)_z \\ 0 & 0 & 0 & n_x & n_y & n_z \\ 0 & 0 & 0 & o_x & o_y & o_z \\ 0 & 0 & 0 & a_x & a_y & a_z \end{bmatrix} \begin{bmatrix} d_x \\ d_y \\ d_z \\ \delta_x \\ \delta_y \\ \delta_z \end{bmatrix}$$



雅克比矩阵构建(第i列)

✓ 对于移动关节，雅克比矩阵的第 i 列计算式为：

$${}^T J_i = \begin{bmatrix} n_z \\ o_z \\ a_z \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} {}^T d_x \\ {}^T d_y \\ {}^T d_z \\ {}^T \delta_x \\ {}^T \delta_y \\ {}^T \delta_z \end{bmatrix} = \begin{bmatrix} n_x & n_y & n_z & (p \times n)_x & (p \times n)_y & (p \times n)_z \\ o_x & o_y & o_z & (p \times o)_x & (p \times o)_y & (p \times o)_z \\ a_x & a_y & a_z & (p \times a)_x & (p \times a)_y & (p \times a)_z \\ 0 & 0 & 0 & n_x & n_y & n_z \\ 0 & 0 & 0 & o_x & o_y & o_z \\ 0 & 0 & 0 & a_x & a_y & a_z \end{bmatrix} \begin{bmatrix} d_x \\ d_y \\ d_z \\ \delta_x \\ \delta_y \\ \delta_z \end{bmatrix}$$



雅克比矩阵构建方法(微分变换法)

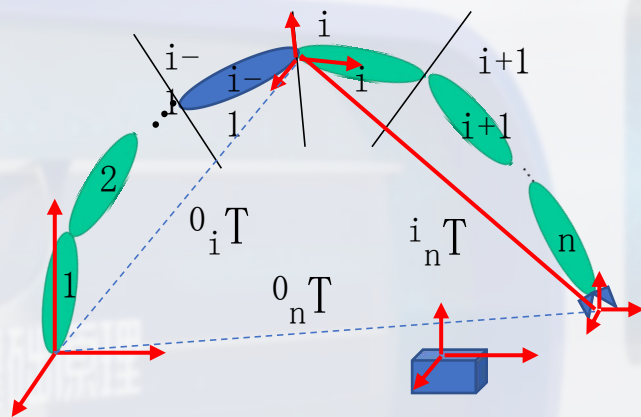
$${}^T J_i = \begin{bmatrix} (P \times n)_z \\ (P \times o)_z \\ (P \times a)_z \\ n_z \\ o_z \\ a_z \end{bmatrix} \quad {}^T J_i = \begin{bmatrix} n_z \\ o_z \\ a_z \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$${}^i_n T = {}^i_{i+1} T \dots {}^{n-1}_n T = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^T J = {}^n J = \begin{bmatrix} {}^T J_1 & {}^T J_2 & \dots & {}^T J_i & \dots & {}^T J_n \end{bmatrix}$$



雅克比矩阵构建方法(微分变换法)



- 计算 ${}^0_1T, {}^1_2T, \dots, {}^{n-1}_nT$
- 计算 ${}^{n-1}_nT, {}^{n-2}_nT, \dots, {}^i_nT, \dots, {}^0_nT$

$${}^{n-1}_nT = {}^{n-1}_nT$$

$${}^{n-2}_nT = {}^{n-2}_{n-1}T {}^{n-1}_nT$$

$$\dots \dots$$

$${}^{i-1}_nT = {}^{i-1}_iT {}^i_nT$$

$$\dots \dots$$

$${}^1_nT = {}^1_2T {}^2_nT$$

$${}^0_nT = {}^0_1T {}^1_nT$$
- 计算J的各列元素，第i列使用 i_nT



雅克比矩阵是相对于末端坐标系构建的

$${}^T J = {}^n J = \begin{bmatrix} {}^T J_1 & {}^T J_2 & \dots & {}^T J_i & \dots & {}^T J_n \end{bmatrix}$$

$${}^0 J = \begin{bmatrix} {}^0_n R & 0 \\ 0 & {}^0_n R \end{bmatrix} {}^n J$$

